Rational Functions and Asymptotes

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Finding asymptotes and intercepts**

For $f(x) = \frac{x+2}{x-3}$, find:

a the vertical asymptote $x = 3$

b the horizontal asymptote $y = 1$

c the x-intercept $(-2, 0)$

d the y-intercept $\left(0, -\frac{2}{3}\right)$

(a) $x = 3$; (b) $y = 1$; (c) $(-2, 0)$; (d) $\left(0, -\frac{2}{3}\right)$

2 **[Foundational]** • **Target: Finding asymptotes and intercepts**

Find all asymptotes of $f(x) = \frac{2x^2+1}{x^2-4}$.

Vertical: $x = 2$ and $x = -2$. Horizontal: equal degrees, so $y = \frac{2}{1} = 2$.

3 **[Foundational]** • **Target: Finding asymptotes and intercepts**

Under the structural constraint that consider $f(x) = \frac{x^2-9}{x-3}$.

a Simplify f and identify the feature at $x = 3$. $f(x) = x + 3$ for $x \neq 3$: a hole at $(3, 6)$.

b Does the graph have a vertical asymptote? No — the discontinuity is removable, not an asymptote.

(a) $f(x) = x + 3$ for $x \neq 3$: a hole at $(3, 6)$.; (b) No — the discontinuity is removable, not an asymptote.

4 **[Foundational]** • **Target: Behaviour near asymptotes**

For $f(x) = \frac{x+2}{x-3}$, describe the behaviour of $f(x)$ as:

a $x \to 3^+$ $f(x) \to +\infty$

b $x \to 3^-$ $f(x) \to -\infty$

c $x \to \infty$ $f(x) \to 1$

(a) $f(x) \to +\infty$; (b) $f(x) \to -\infty$; (c) $f(x) \to 1$

5 **[Foundational]** • **Target: Behaviour near asymptotes**

Write a rational function with vertical asymptotes at $x = 1$ and $x = -4$ and horizontal asymptote $y = 0$.

One example: $f(x) = \frac{1}{(x-1)(x+4)}$ (any numerator of degree less than 2 works).

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Classifying end behaviour from degree

Without dividing, state the horizontal or slant asymptote (or neither) for each function based on the degrees of numerator and denominator:

a $f(x) = \frac{3x+1}{x^2-4}$ (degree 1 over 2) Degree of numerator $<$ denominator: horizontal asymptote $y = 0$.

b $f(x) = \frac{5x^2-1}{2x^2+3}$ (degree 2 over 2) Equal degrees: horizontal asymptote $y = \frac{5}{2}$.

c $f(x) = \frac{x^3+2}{x^2-1}$ (degree 3 over 2) Numerator degree is exactly one more: slant asymptote (no horizontal asymptote).

(a) Degree of numerator $<$ denominator: horizontal asymptote $y = 0$.; (b) Equal degrees: horizontal asymptote $y = \frac{5}{2}$.; (c) Numerator degree is exactly one more: slant asymptote (no horizontal asymptote).

7 [Foundational] • Target: Classifying end behaviour from degree

Find the slant asymptote of $f(x) = \frac{2x^2-3x+1}{x+2}$.

Dividing: $2x^2 - 3x + 1 = (x+2)(2x-7) + 15$, so $f(x) = 2x - 7 + \frac{15}{x+2}$. Slant asymptote: $y = 2x - 7$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Holes versus asymptotes

Construct a rational function that has a hole at $x = 3$ and a vertical asymptote at $x = -1$.

One example: $f(x) = \frac{(x-3)(x+2)}{(x-3)(x+1)}$, which simplifies to $\frac{x+2}{x+1}$ with a hole at $x = 3$ and vertical asymptote at $x = -1$.

9 [Intermediate] • Target: Reading asymptotic behavior from a graph

The graph of $f(x) = \frac{1}{x-1}$ is shown. Use it to describe the behavior near $x = 1$ and as $x \to \pm\infty$.

As $x \to 1^-$, $f(x) \to -\infty$; as $x \to 1^+$, $f(x) \to +\infty$ — a vertical asymptote at $x = 1$. As $x \to \pm\infty$, $f(x) \to 0$ — a horizontal asymptote at $y = 0$.

10 [Intermediate] • Target: Domain restrictions and sign analysis

Find the domain of $f(x) = \frac{x-1}{x^2-x-6}$, and determine the sign of $f(x)$ on the interval $(3, \infty)$.

Factor denominator: $x^2 - x - 6 = (x-3)(x+2)$. Domain: $x \neq 3, -2$. On $(3, \infty)$: numerator $x - 1 > 0$, denominator $(x-3)(x+2) > 0$ (both factors positive), so $f(x) > 0$ on this interval.

© ClipSAT — clipsat.org | All Rights Reserved

11 [Intermediate] • Target: Asymptotes in applied contexts

The average cost per unit of producing x items is $C(x) = \frac{500 + 3x}{x}$ (setup cost 500 riyals plus 3 riyals per unit). Find the horizontal asymptote and explain its meaning in context.

As $x \to \infty$, $C(x) \to 3$ (horizontal asymptote $y = 3$). This means as production volume grows very large, the average cost per unit approaches 3 riyals — the fixed setup cost gets spread so thin it becomes negligible.

12 [Intermediate] • Target: Asymptotes in applied contexts

A drug's concentration in the bloodstream is modeled by $C(t) = \frac{5t}{t^2 + 1}$ (mg/L, hours). Find the horizontal asymptote and interpret it, and find $C(0)$.

Since the denominator has higher degree, the horizontal asymptote is $y = 0$ — the concentration eventually approaches zero as the drug is fully metabolized. $C(0) = \frac{0}{1} = 0$ mg/L (none in the bloodstream initially, before administration takes effect).

13 [Intermediate] • Target: Finding a rational function's equation from its features

Find the equation of a rational function with vertical asymptotes at $x = -2$ and $x = 3$, horizontal asymptote $y = 0$, and x-intercept at $x = 1$.

Numerator degree less than denominator (for $y = 0$), zero at $x = 1$: $f(x) = \frac{x - 1}{(x + 2)(x - 3)}$.

14 [Intermediate] • Target: Domain restrictions and sign analysis

For the rational function $r(x) = \frac{2x^2 - 2x - 12}{x^2 - 9}$, find: (a) the domain, (b) the coordinates of any removable discontinuity (hole), (c) the vertical and horizontal asymptotes, and (d) the x- and y-intercepts.

Step 1: Factor numerator and denominator completely: $r(x) = \frac{2(x^2 - x - 6)}{(x - 3)(x + 3)} = \frac{2(x - 3)(x + 2)}{(x - 3)(x + 3)}$. Step 2: Answer each part:(a) Domain: $(-\infty, -3) \cup (-3, 3) \cup (3, \infty)$, i.e. $x \neq \pm 3$.(b) Removable Discontinuity: The factor $(x - 3)$ cancels; for $x \neq 3$, $r(x) = \frac{2(x + 2)}{x + 3}$. Evaluating at $x = 3$ gives $y = \frac{2(5)}{6} = \frac{5}{3}$. Hole at $\left(\mathbf{3, \frac{5}{3}}\right)$.(c) Asymptotes: Vertical asymptote at $\mathbf{x = -3}$; ho.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Asymptotes in applied contexts • Cross-Lesson: 3.03 Sum and Difference Formulas

Level 1 · FoundationsLet $q(x) = -(x + 2)^2(x - 1)(x - 4)^3$. What is the degree of $q(x)$, does the graph cross or touch the x-axis at $x = -2$, and what is $\lim_{x \to \infty} q(x)$?

Sum the exponents of the linear factors to find the degree: $2 + 1 + 3 = 6 \Rightarrow \text{degree } 6$. At $x = -2$, the multiplicity is 2 (even), so the graph touches the x-axis and turns around without crossing. The degree is even (6) and the leading coefficient is negative (-1), so $\lim_{x \to \infty} q(x) = -\infty$.

© ClipSAT — clipsat.org | All Rights Reserved

16 **[Challenging]** • **Target: Finding a rational function's equation from its features** • **Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations**

Let $f(x) = \frac{2x+1}{x-3}$ for $x \neq 3$. (a) Determine the inverse function $f^{-1}(x)$. (b) State the domain and range of $f^{-1}(x)$.

(a) Find the inverse function: Set $y = \frac{2x+1}{x-3}$. Swap x and y: $x = \frac{2y+1}{y-3} \Rightarrow x(y-3) = 2y+1 \Rightarrow xy - 3x = 2y + 1$. Group y terms: $xy - 2y = 3x + 1 \Rightarrow y(x-2) = 3x + 1 \Rightarrow \mathbf{f^{-1}(x) = \frac{3x+1}{x-2}}$. (b) Domain and Range: The denominator of $f^{-1}(x)$ cannot be zero, so the domain is $\mathbf{(-\infty, 2) \cup (2, \infty)}$ (all $x \neq 2$). The range of f^{-1} equals the domain of f, which is.

17 **[Challenging]** • **Target: Finding asymptotes and intercepts** • **Cross-Lesson: 4.01 Circles and Parabolas**

Let $f(x) = \frac{x^2 - 9}{x^2 - 5x + 6}$. Identify the locations of all vertical asymptotes and removable discontinuities (holes) of $f(x)$.

Factor numerator and denominator completely: $f(x) = \frac{(x-3)(x+3)}{(x-2)(x-3)}$ Notice that $(x-3)$ is a common factor in both numerator and denominator: Removable Discontinuity (Hole): Occurs at $x = 3$. To find the y -coordinate, evaluate the simplified fraction: $f_{\text{reduced}}(3) = \frac{3+3}{3-2} = \frac{6}{1} = 6$. The hole is at **(3, 6)**.Vertical Asymptote: The remaining uncancelled denominator factor is $x - 2 = 0$, creating a vertical asymptote at line $\mathbf{x = 2}$.

18 **[Challenging]** • **Target: Behaviour near asymptotes** • **Cross-Lesson: 4.02 Ellipses**

The function f is defined by $f(x) = x^2 - 6x + 13$ for $x \geq 3$. (a) State the range of f. (b) Find an expression for $f^{-1}(x)$ and state the domain and range of f^{-1}.

(a) Complete the square: $f(x) = (x-3)^2 - 9 + 13 = (x-3)^2 + 4$. Since $x \geq 3$, $(x-3)^2 \geq 0$, so the minimum value is $f(3) = 4$. The range of f is $f(x) \geq 4$. (b) Set $y = (x-3)^2 + 4$. Rearrange for x: $y - 4 = (x-3)^2 \Rightarrow x - 3 = +\sqrt{y-4}$ (since $x \geq 3$) $x = 3 + \sqrt{y-4}$ Therefore, $f^{-1}(x) = 3 + \sqrt{x-4}$. Domain of f^{-1} is $x \geq 4$ (the range of f), and range of f^{-1} is $f^{-1}(x) \geq 3$ (the domain of f).

19 **[Challenging]** • **Target: Classifying end behaviour from degree** • **Cross-Lesson: 4.03 Hyperbolas**

Find the equation of the slant (oblique) asymptote of the rational function $g(x) = \frac{3x^2 - 5x + 7}{x - 2}$, and compute $\lim_{x \to 2^+} g(x)$.

Step 1: Perform polynomial long division (or synthetic division by $c = 2$): $3x^2 - 5x + 7 = (x-2)(3x+1) + 9 \Rightarrow g(x) = 3x + 1 + \frac{9}{x-2}$. Step 2: Identify the slant asymptote: As $x \to \pm\infty$, $\frac{9}{x-2} \to 0$, so the slant asymptote is $\mathbf{y = 3x + 1}$. Step 3: One-sided limit at $x = 2^+$: As $x \to 2^+$, $x - 2 \to 0^+$ and numerator $\to 9 > 0$, so $\lim_{x \to 2^+} g(x) = \mathbf{+\infty}$.

© ClipSAT — clipsat.org | All Rights Reserved

20 [Challenging] • Target: Holes versus asymptotes • Cross-Lesson: 5.01 Vectors: Components and Operations

Let $f(x) = \frac{4e^{2x}}{e^{2x}+3}$. Prove that f is strictly one-to-one and find an explicit formula for its inverse function $f^{-1}(x)$, stating the domain of f^{-1}.

Step 1: Solve $y = \frac{4e^{2x}}{e^{2x}+3}$ for e^{2x}: $y(e^{2x}+3) = 4e^{2x} \Rightarrow 3y = e^{2x}(4-y) \Rightarrow e^{2x} = \frac{3y}{4-y}$. Step 2: Take the natural logarithm and divide by 2: $2x = \ln\left(\frac{3y}{4-y}\right) \Rightarrow f^{-1}(x) = \mathbf{\frac{1}{2}}\ln\left(\mathbf{\frac{3x}{4-x}}\right)$. Step 3: Domain of f^{-1} (range of f): Since $e^{2x} > 0$, we require $\frac{3x}{4-x} > 0 \Rightarrow \mathbf{0 < x < 4}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Finding asymptotes and intercepts

Consider the quadratic function $f(x) = x^2 - 4x + 3$. What are the x-intercepts of the graph of this function?

a. $(1, 0)$ **and** $(3, 0)$ ✓

b. $(-1, 0)$ and $(-3, 0)$

c. $(1, 0)$ and $(-3, 0)$

d. $(-1, 0)$ and $(3, 0)$

To find the x-intercepts of the graph of a function, we need to find the values of x for which $f(x) = 0$. For the given function $f(x) = x^2 - 4x + 3$, we set $x^2 - 4x + 3 = 0$. This is a quadratic equation that can be solved by factoring. We look for two numbers that multiply to 3 and add to -4. These numbers are -1 and -3. Thus, we can factor the equation as $(x - 1)(x - 3) = 0$. Setting each factor to zero, we get $x - 1 = 0$ or $x - 3 = 0$. Solving these equations gives $x = 1$ and $x = 3$. Therefore, the x.

22 [Foundational] • Target: Behaviour near asymptotes

Consider the quadratic function $f(x) = x^2 - 6x + 5$. What are the x-intercepts of the graph of $f(x)$?

a. **x = 1 and x = 5** ✓

b. x = -1 and x = -5

c. x = 1 and x = -5

d. x = -1 and x = 5

To find the x-intercepts, we set $f(x) = 0$. Thus, we need to solve the equation $x^2 - 6x + 5 = 0$. This is a quadratic equation that can be factored. We look for two numbers that multiply to 5 and add to -6. These numbers are -1 and -5. Therefore, the equation can be factored as $(x - 1)(x - 5) = 0$. Setting each factor to zero, we get $x - 1 = 0$ or $x - 5 = 0$. Solving for x, we find $x = 1$ or $x = 5$. Thus, the x-intercepts are at $x = 1$ and $x = 5$.

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Classifying end behaviour from degree

The graph of a polynomial function $f(x)$ of degree 4 is shown in the figure below. The graph is tangent to the x-axis at $(-2, 0)$, intersects the x-axis at $(1, 0)$ and $(3, 0)$, and crosses the y-axis at $(0, -6)$. A new polynomial $Q(x)$ is defined by $Q(x) = f(x) + \frac{1}{2}x^3$. When $Q(x)$ is divided by $x^2 - 4$, the remainder is the linear polynomial $R(x)$. What is the value of $R(5)$?

a. 24 ✓

b. 18

c. 17

d. 14

We solve the problem in two key phases: identifying the polynomial $f(x)$ from its graphical features, and applying the Polynomial Remainder Theorem to determine $R(x)$. **Step 1: Determine $f(x)$** - Because the graph is tangent to the x-axis at $x = -2$, $(x + 2)$ is a factor with an even multiplicity of at least 2. - Because the graph crosses the x-axis at $x = 1$ and $x = 3$, $(x - 1)$ and $(x - 3)$ are factors with odd multiplicities of at least 1. - Since $f(x)$ is of degree 4, the sum of multipliciti.

24 [Foundational] • Target: Holes versus asymptotes

The graph of a polynomial function $f(x)$ is shown below. Which of the following could be the equation for $f(x)$?

a. $f(x) = (x + 2)(x - 1)(x - 3)$ ✓

b. $f(x) = (x - 2)(x + 1)(x + 3)$

c. $f(x) = -(x + 2)(x - 1)(x - 3)$

d. $f(x) = (x + 2)(x - 1)^2(x - 3)$

To determine which polynomial function matches the graph, we analyze the intercepts, multiplicities, and end behavior: 1. **Identify the x-intercepts (zeros):** The graph crosses the x-axis at $x = -2$, $x = 1$, and $x = 3$. By the Factor Theorem, if c is a zero of $f(x)$, then $(x - c)$ is a factor. Therefore, the corresponding linear factors are: $(x - (-2)) = (x + 2), \quad (x - 1), \quad \text{and} \quad (x - 3)$ 2. **Multiplicity:** At each of the three x -intercepts, the graph passes directly through.

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Reading asymptotic behavior from a graph

Which of the following statements describes the end behavior of the polynomial function $f(x) = -5x^4 + 2x^3 - 7x + 3$?

a. **As $x \to -\infty$, $f(x) \to -\infty$, and as $x \to \infty$, $f(x) \to -\infty$** ✓

b. As $x \to -\infty$, $f(x) \to \infty$, and as $x \to \infty$, $f(x) \to -\infty$

c. As $x \to -\infty$, $f(x) \to -\infty$, and as $x \to \infty$, $f(x) \to \infty$

d. As $x \to -\infty$, $f(x) \to \infty$, and as $x \to \infty$, $f(x) \to \infty$

To determine the end behavior of a polynomial function, examine its leading term, which dominates the function's values for very large positive and negative values of x. 1. Identify the leading term: The highest power of x is 4, so the leading term is $-5x^4$. 2. Analyze the degree and leading coefficient: - The degree is $n = 4$, which is an even integer. - The leading coefficient is $a = -5$, which is negative. 3. Determine the limits as $x \to \pm\infty$: - As $x \to \infty$.

26 [Foundational] • Target: Domain restrictions and sign analysis

For the polynomial $f(x) = 3x^2 + 2x - 1$, describe the end behavior of f.

a. down on the left, up on the right (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to +\infty$)

b. down on both ends (as $x \to -\infty$, $f(x) \to -\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

c. up on the left, down on the right (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$)

d. **up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$)** ✓

The leading term is $3x^2$, degree even, leading coefficient positive, so the end behavior is: up on both ends (as $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to +\infty$).

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Asymptotes in applied contexts

Find the x-intercepts and vertical asymptotes of $f(x) = \frac{x^2 - x - 6}{x^2 - 4}$.

a. x-int: $x = 3,\ -2$ (but $x = -2$ is a hole); VA: $x = 2$

b. x-int: $x = 3$; VA: $x = \pm 2$

c. x-int: $x = 3,\ -2$; VA: $x = \pm 2$

d. **x-int:** $x = 3$**; VA:** $x = 2$ **only** ✓

Num: $(x - 3)(x + 2)$. Den: $(x - 2)(x + 2)$. Cancel $(x + 2)$: hole at $x = -2$. VA at $x = 2$. x-int: set num of simplified function $= 0$: $x = 3$.

28 [Foundational] • Target: Finding a rational function's equation from its features

Find the inverse of $f(x) = \frac{3x + 2}{x - 1}$ and state its domain.

a. $f^{-1}(x) = \frac{x + 2}{x - 3}$**; domain: all reals except** $x = 3$ ✓

b. $f^{-1}(x) = \frac{3x - 2}{x + 1}$

c. $f^{-1}(x) = \frac{x + 2}{x + 3}$

d. $f^{-1}(x) = \frac{x - 2}{x + 3}$

Swap: $x = \frac{3y + 2}{y - 1} \Rightarrow x(y - 1) = 3y + 2 \Rightarrow xy - x = 3y + 2 \Rightarrow y(x - 3) = x + 2 \Rightarrow y = \frac{x + 2}{x - 3}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Finding asymptotes and intercepts

Find all intercepts and asymptotes of $f(x) = \frac{x^2 - x - 6}{x^2 - 4}$.

a. x-int $x = 3$; VA $x = 2$; slant asymptote

b. Only HA $y = 1$

c. x-int $x = 3,\ -2$; VA $x = \pm 2$; HA $y = 0$

d. **x-int** $x = 3$**; hole at** $x = -2$**; VA** $x = 2$**; HA** $y = 1$ ✓

Simplified: $\frac{x - 3}{x - 2}$. HA $y = 1$; VA $x = 2$; hole $x = -2$; x-int $x = 3$.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Behaviour near asymptotes

The reciprocal function $f(x) = \frac{1}{x}$ has asymptotes:

a. $x = 0$ **and** $y = 0$ ✓

b. No asymptotes

c. $x = -1$ and $y = -1$

d. $x = 1$ and $y = 1$

Classic reciprocal function: both axes are asymptotes.

31 [Intermediate] • Target: Classifying end behaviour from degree

A polynomial of even degree with positive leading coefficient has end behavior:

a. $y \to -\infty$ as $x \to \pm\infty$

b. $y \to \infty$ as $x \to +\infty$ and $y \to -\infty$ as $x \to -\infty$

c. $y \to +\infty$ **as** $x \to \pm\infty$ ✓

d. Constant

Even degree, positive leading coeff: up on both ends.

32 [Intermediate] • Target: Holes versus asymptotes

What causes a hole in the graph of a rational function?

a. The y-intercept

b. A horizontal asymptote

c. **A common factor in numerator and denominator** ✓

d. A vertical asymptote

Removable discontinuity from cancellation.

33 [Intermediate] • Target: Reading asymptotic behavior from a graph

If a rational function has no real zeros in its denominator, it has:

a. No horizontal asymptote

b. No x-intercepts

c. **No vertical asymptotes** ✓

d. No range

VA occur where denominator is zero.

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: Domain restrictions and sign analysis

Consider the polynomial function $f(x) = x^2 - 5x + 6$. What are the roots of the function $f(x)$?

a. 1 and 2

b. **2 and 3 ✓**

c. 3 and 4

d. 1 and 6

To find the roots of the function $f(x) = x^2 - 5x + 6$, we need to solve the equation $f(x) = 0$. This means we need to find the values of x for which $x^2 - 5x + 6 = 0$. This is a quadratic equation that can be solved by factoring. We are looking for two numbers that multiply to 6 and add up to -5. These numbers are -2 and -3. Therefore, we can factor the quadratic as $(x - 2)(x - 3) = 0$. Setting each factor to zero gives us the roots: $x - 2 = 0 \Rightarrow x = 2$ and $x - 3 = 0 \Rightarrow x = 3$. Thus, the roots of.

35 [Intermediate] • Target: Asymptotes in applied contexts

The graph of the rational function $g(x) = \frac{ax^2 + bx + 1}{x^2 - 4}$ has a horizontal asymptote at $y = 2$ and a vertical asymptote at $x = 2$. Given that $g(0) = -1/4$, find the value of a.

a. 1

b. **2 ✓**

c. 3

d. 4

The horizontal asymptote of a rational function $g(x) = \frac{P(x)}{Q(x)}$ where the degree of $P(x)$ is equal to the degree of $Q(x)$ is the ratio of the leading coefficients. In this case, the leading coefficient of the numerator is a and the leading coefficient of the denominator is 1. Therefore, the horizontal asymptote is $y = \frac{a}{1} = a$. We are given that the horizontal asymptote is $y = 2$, so $a = 2$. The vertical asymptotes occur at the zeros of the denominator that are not zeros of the numerator.

36 [Intermediate] • Target: Finding a rational function's equation from its features

The function f is defined by $f(x) = \frac{ax + 8}{cx + d}$ for all real numbers $xeq -\frac{d}{c}$, where a, c, and d are nonzero constants. In the standard (x, y) coordinate plane, the graph of $y = f(x)$ has a horizontal asymptote at $y = 2$, a vertical asymptote at $x = 3$, and an x-intercept at $(-4, 0)$. What is the value of $f^{-1}(4)$?

a. **10 ✓**

b. 16

c. -2

d. 2

We determine the constants a, c, and d step-by-step: 1. **Use the x-intercept:** The point $(-4, 0)$ lies on the graph, meaning $f(-4) = 0$. For a rational function, the numerator must equal zero at an x-intercept: $a(-4) + 8 = 0 \Rightarrow -4a = -8 \Rightarrow a = 2$ 2. **Use the horizontal asymptote:** For $f(x) = \frac{ax + 8}{cx + d}$, as $x \to \pm\infty$, the horizontal asymptote is given by the ratio of the leading coefficients: $y = \frac{a}{c} = 2$ Since $a = 2$, we have.

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Finding asymptotes and intercepts

The diagram shows the graph of the function $y = f(x)$. The function $f(x)$ is defined for all real numbers. Which of the following represents the graph of $y = f^{-1}(x)$?

a. Graph A

b. **Graph B ✓**

c. Graph C

d. Graph D

The graph of the inverse function $y = f^{-1}(x)$ is obtained by reflecting the graph of $y = f(x)$ across the line $y = x$. The provided graph of $f(x)$ is a parabola opening upwards with its vertex at $(2, -1)$. Graph A shows a function that is not a reflection of $f(x)$ across $y = x$. Graph B correctly shows the reflection of $y = f(x)$ across the line $y = x$. For example, the point $(0, 3)$ on $f(x)$ corresponds to the point $(3, 0)$ on $f^{-1}(x)$. The vertex $(2, -1)$ on $f(x)$ corresponds to the point.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Behaviour near asymptotes • Cross-Lesson: 4.02 Ellipses

In the standard (x, y) coordinate plane, what is the equation of the horizontal asymptote of the graph of the function $f(x) = \frac{8x+3}{2x-5}$?

a. $y = 4$ ✓

b. $y = \frac{5}{2}$

c. $x = \frac{5}{2}$

d. $y = -\frac{3}{5}$

To find the horizontal asymptote of a rational function $f(x) = \frac{P(x)}{Q(x)}$ where the degrees of the numerator and denominator are equal, divide the leading coefficient of the numerator by the leading coefficient of the denominator: $y = \lim_{x \to \pm\infty} \frac{8x+3}{2x-5} = \lim_{x \to \pm\infty} \frac{8+\frac{3}{x}}{2-\frac{5}{x}} = \frac{8}{2} = 4$ Therefore, the horizontal asymptote is the line $y = 4$. Misconception checks: - $x = \frac{5}{2}$ is the vertical asymptote, foun.

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Classifying end behaviour from degree • Cross-Lesson: 4.03 Hyperbolas

For the rational function $f(x) = \frac{x-2}{x+3}$, what value of x is excluded from the domain?

a. x = -3 ✓

b. x = 2

c. x = 0

d. x = -2

A rational function is undefined when its denominator is equal to zero. For the function $f(x) = \frac{x-2}{x+3}$, the denominator is $x + 3$. To find the excluded value from the domain, we set the denominator equal to zero and solve for x: $x + 3 = 0$. Subtracting 3 from both sides gives $x = -3$. Therefore, $x = -3$ is excluded from the domain.

40 [Challenging] • Target: Holes versus asymptotes • Cross-Lesson: 5.01 Vectors: Components and Operations

Consider the polynomial function $f(x) = x^3 - 5x^2 + 2x + 8$. What is the value of $f(2)$?

a. -4 ✓

b. 0

c. 4

d. 8

To find the value of $f(2)$, substitute $x = 2$ into the polynomial function $f(x) = x^3 - 5x^2 + 2x + 8$. $f(2) = (2)^3 - 5(2)^2 + 2(2) + 8$ $f(2) = 8 - 5(4) + 4 + 8$ $f(2) = 8 - 20 + 4 + 8$ $f(2) = -12 + 4 + 8$ $f(2) = -8 + 8$ $f(2) = 0$.

41 [Challenging] • Target: Reading asymptotic behavior from a graph • Cross-Lesson: 5.02 The Dot Product

Determine if $f(x) = \frac{3x^2 - 5}{x^2 + 2}$ has any vertical asymptotes.

a. Yes at $x = 0$

b. Yes at $x = \pm\sqrt{2}$

c. Yes at $y = 3$

d. No (denominator $x^2 + 2 > 0$ always) ✓

Applying Rational Expressions & Functions principles, $x^2 + 2 \geq 2 > 0$: no real zeros. Therefore, the verified correct option is No (denominator $x^2 + 2 > 0$ always).

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Domain restrictions and sign analysis • Cross-Lesson: 5.03 Parametric Equations

Find the cubic polynomial $P(x)$ with leading coefficient 2 whose graph passes through $(0, 6)$ and has zeros $1, -3$ and $1 + i$. But wait — to have real coefficients, if $1 + i$ is a root, so is $1 - i$. That gives degree 4.

a. $P(x) = 2x^3 - 6$

b. $P(x) = 2(x - 1)(x + 3)(x - 1 - i)$

c. $P(x) = 2(x - 1)^2(x + 3)$

d. **This is impossible for a cubic with real coefficients (need degree ≥4 for those zeros)** ✓

Complex roots of polynomials with real coefficients come in conjugate pairs, so $1 + i$ forces $1 - i$ as well → at least degree 4. Impossible for cubic.

43 [Challenging] • Target: Asymptotes in applied contexts • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Which of these is a rational function?

a. $f(x) = \sqrt{x}$

b. $f(x) = \frac{x^2 - 1}{x + 3}$ ✓

c. $f(x) = x^{2/3}$

d. $f(x) = 2^x$

Applying Rational Expressions & Functions principles, Ratio of polynomials only. Therefore, the verified correct option is $f(x) = \frac{x^2 - 1}{x + 3}$.

44 [Challenging] • Target: Finding a rational function's equation from its features • Cross-Lesson: 6.02 Polar Equations and Graphs

Find the vertical asymptotes of $f(x) = \frac{x + 1}{x^2 - 4}$.

a. $x = \pm 2$ ✓

b. No VA

c. $x = -1, \pm 2$

d. $x = 2$ only

Applying Rational Expressions & Functions principles, $x^2 - 4 = 0 \Rightarrow x = \pm 2$. Therefore, the verified correct option is $x = \pm 2$.

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Finding asymptotes and intercepts • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

The domain of $f(x) = \frac{1}{x-3}$ excludes:

a. $x = 3$ ✓

b. $x = 0$

c. All reals

d. $x = -3$

Applying Rational Expressions & Functions principles, Denominator cannot be zero. Therefore, the verified correct option is $x = 3$.

© ClipSAT — clipsat.org | All Rights Reserved